

```
// Experiment 1: Introduction to Arduino and Digital I/O Control
```

```
int push1Pin = 5;  
int push2Pin = 4;  
int push3Pin = 2;  
int push4Pin = 3;
```

```
int push1Value;  
int push2Value;  
int push3Value;  
int push4Value;
```

```
int ledPin1 = 7;  
int ledPin2 = 8;  
int ledPin3 = 9;  
int ledPin4 = 10;
```

```
void setup() {  
    Serial.begin(9600);  
  
    pinMode(push1Pin, INPUT);  
    pinMode(push2Pin, INPUT);  
    pinMode(push3Pin, INPUT);  
    pinMode(push4Pin, INPUT);  
  
    pinMode(ledPin1, OUTPUT);
```

```
pinMode(ledPin2, OUTPUT);
pinMode(ledPin3, OUTPUT);
pinMode(ledPin4, OUTPUT);

Serial.println("System Initialized: Monitoring Push Buttons and
LEDs...");
}

void loop() {
    push1Value = digitalRead(push1Pin);
    push2Value = digitalRead(push2Pin);
    push3Value = digitalRead(push3Pin);
    push4Value = digitalRead(push4Pin);

    if(push1Value == HIGH) {
        digitalWrite(ledPin1, HIGH);
        digitalWrite(ledPin2, LOW);
        digitalWrite(ledPin3, LOW);
        digitalWrite(ledPin4, LOW);
        Serial.println("Button1 PRESSED → LED1 ON");
    }
    else if(push2Value == HIGH) {
        digitalWrite(ledPin1, LOW);
        digitalWrite(ledPin2, HIGH);
        digitalWrite(ledPin3, LOW);
        digitalWrite(ledPin4, LOW);
    }
}
```

```
        Serial.println("Button2 PRESSED → LED2 ON");
    }
    else if(push3Value == HIGH) {
        digitalWrite(ledPin1, LOW);
        digitalWrite(ledPin2, LOW);
        digitalWrite(ledPin3, HIGH);
        digitalWrite(ledPin4, LOW);
        Serial.println("Button3 PRESSED → LED3 ON");
    }
    else if(push4Value == HIGH) {
        digitalWrite(ledPin1, LOW);
        digitalWrite(ledPin2, LOW);
        digitalWrite(ledPin3, LOW);
        digitalWrite(ledPin4, HIGH);
        Serial.println("Button4 PRESSED → LED4 ON");
    }
    else {
        digitalWrite(ledPin1, LOW);
        digitalWrite(ledPin2, LOW);
        digitalWrite(ledPin3, LOW);
        digitalWrite(ledPin4, LOW);
        Serial.println("No button pressed → All LEDs OFF");
    }

    Serial.print("Buttons: ");
    Serial.print(push1Value);
```

```
Serial.print(" ");  
Serial.print(push2Value);  
Serial.print(" ");  
Serial.print(push3Value);  
Serial.print(" ");  
Serial.println(push4Value);  
  
delay(500);  
}
```